



SCHOOL OF COMPUTING AND INFORMATICS

DEPARTMENT OF COMPUTING AND INFORMATICS

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Contents

1. Introduction.....	3
2. Objectives	3
3. Tools and Technologies Used.....	3
4. Week 1 Activities: Environment Setup & Foundation App	4
5. Week 2 Activities: Mini Student Management System	5
6. Week 3 Activities: Pharmacy Management UI Prototype	6
7. Challenges Encountered.....	6
8. Solutions Applied.....	7
9. Conclusion	7

1. Introduction

The purpose of this project was to progressively learn and implement mobile application development concepts across three distinct phases. The project began with environment configuration and the creation of a foundational Flutter application. It then advanced to developing a functional, locally-stored Mini Student Management System using Dart and Flutter. Finally, the project culminated in designing a comprehensive UI/UX prototype for a Pharmacy Management System using Figma, which was subsequently translated into a fully functional, mobile-first web prototype using React and modern frontend technologies.

2. Objectives

- Install and configure essential development tools including Git, Android Studio, and the Flutter SDK.
- Create foundational Flutter applications and customize UI elements such as app titles, background colors, and typography.
- Build functional authentication and registration forms with validation logic.
- Implement local data persistence to save and retrieve application records.
- Design a responsive, modern UI prototype containing a minimum of 5 interconnected screens.

3. Tools and Technologies Used

Tool/Technology	Purpose in Project
Flutter & Dart	Primary framework and language for developing the Week 1 application and the Week 2 Student Management System.

Tool/Technology	Purpose in Project
Android Studio	The integrated development environment (IDE) configured for mobile app testing and development.
Git	Version control system, installed with file system caching enabled for performance.
Figma	Used in Week 3 to design the initial high-fidelity wireframes and interaction flows.
React & TypeScript	Frontend framework used to build the interactive UI prototype for the Pharmacy app.
React Router v7	Handled navigation between the 7 different screens in the React prototype.
Tailwind CSS v4	Utilized for responsive, utility-first styling across the React application.
Motion (Framer)	Implemented for smooth page transitions and UI element animations.

4. Week 1 Activities: Environment Setup & Foundation App

The first week was dedicated to establishing the development environment and verifying the toolchain.

- Installation Phase: Successfully installed Git and configured Android Studio with the necessary virtual devices for testing.

- Application Initialization: Utilized the command line to generate a new project named `assignment_one`.
- UI Customization: Modified `main.dart` to meet specific assignment criteria:
 - App Title: Updated metadata and visible headers to display "Assignment One" and "My First Flutter App".
 - Background Color: Customized the Scaffold background to a specific Light Teal (`0xFFE0F2F1`).
 - Typography & Interactions: Rendered "Hello World!" with an updated font size of 35.0 and bold weight. Added an `ElevatedButton` with an icon that successfully triggers a `SnackBar` displaying "Button Clicked Successfully!" upon interaction.
- Visual Evidence: Screenshots of the Git installer, Android Studio setup, terminal execution, and the final rendering in a Chrome debug environment are included in the logbook.

5. Week 2 Activities: Mini Student Management System

The second assignment focused on application logic, structuring a mini student system using a "modular clean-architecture pattern".

- Architecture Implementation: The codebase was separated into distinct layers:
 - `lib/models/student.dart` for the data blueprint.
 - `lib/database/mock_db.dart` to act as an in-memory local data repository.
 - `lib/screens/` containing discrete UI views (`login_page.dart`, `register_page.dart`, `dashboard_page.dart`).
- Registration & Validation Logic: Implemented form validation that checks the `globalStudents` array to prevent duplicate registrations using the Registration Number (`s.regNo == regCtrl.text`). Valid entries are appended to local state.
- Dashboard Rendering: Created a modern profile dashboard that dynamically lists active registered students, displaying their name, registration number, and course.

- Visual Evidence: Screenshots of the centralized login card, the student registration form, and the populated active student dashboard are documented.

6. Week 3 Activities: Pharmacy Management UI Prototype

The final week required the design and development of a modern, multi-screen UI prototype, specifically building the "Gathimaini Pharmacy" application.

- Figma Wireframing: Designed high-fidelity screens utilizing a healthcare green design system (#1B8F4A), Poppins typography, and soft UI shadows. Mapped out user flows connecting the Login, Registration, Dashboard, Upload, and Profile screens.
- React Frontend Engineering: Transitioned the design into code, developing a 7-screen interactive application .
- Key Features Developed:
 - Mobile-First Wrapper: Created a custom PhoneFrame component with a notch and home indicator to simulate a native mobile viewport inside the browser .
 - Bottom Navigation: Implemented persistent routing tabs for Home, Prescription, Cart, and Profile .
 - Cart System: Engineered a functional cart screen that tracks item quantities, removes items, and dynamically calculates subtotals alongside a fixed KES 150 delivery fee to generate a final total .
- Visual Evidence: Comprehensive Figma workspace screenshots and the React component structures are included in the source logbook .

7. Challenges Encountered

- Local State Persistence: Simulating database operations in Week 2 using only local memory arrays meant that application state was volatile and would reset during hot restarts, making consistent testing challenging.

- **Component Routing:** Implementing a seamless, native-feeling navigation experience across 7 distinct screens in the Week 3 React prototype required complex state and layout management.
- **Viewport Consistency:** Ensuring the web-based React prototype accurately mimicked a native mobile application environment without visual distortion on larger desktop screens.

8. Solutions Applied

- **Local State Persistence:** Separated the mock database (`mock_db.dart`) entirely from the UI logic to ensure clean separation of concerns, allowing for reliable simulated data injection during the application's runtime.
- **Component Routing:** Utilized React Router v7 mapped through a central `routes.tsx` file to strictly define paths, coupled with Motion to add subtle layout transitions between screen changes .
- **Viewport Consistency:** Engineered a fixed 375x812 pixel PhoneFrame component to act as an absolute constraint, forcing the inner application content to scroll and behave exactly as it would on a physical mobile device .

9. Conclusion

This iterative project successfully demonstrated the lifecycle of mobile application development. Starting from basic environment setup and UI modifications in Flutter, the project evolved into handling structural architecture and local state management for a student registry. The final phase synthesized these concepts into a highly polished, interactive React prototype for a pharmacy system, ultimately providing comprehensive experience in modern UI design, state calculation, and component-based routing architectures.